FinSight

Relational Schema Diagram

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Exact implemented names | PK dependencies in red | FK references in black | PostgreSQL 17 + MongoDB

app_user

user_id	email	password_hash	display_name	role	is_active	last_login_at	created_at

customer

customer_id	user_id	full_name	date_of_birth	email	phone	country_code	status	kyc_verified_on	updated_at

address

address_id	customer_id	line1	city	state	postal_code	country_code	latitude	longitude	is_primary

account

account_id	customer_id	account_number	account_type	currency_code	balance	status	opened_at	updated_at

payment_method

method_id	account_id	method_type	token	last4	expiry_date	status	updated_at

merchant

merchant_id	merchant_name	category_code	country_code	city	latitude	longitude	risk_tier	is_watchlisted	updated_at

device_fingerprint

device_id	device_hash	os_family	first_seen_country	first_seen_at

transaction

transaction_id	account_id	method_id	merchant_id	device_id	amount	currency_code	exchange_rate	amount_base	direction	txn_type	status	occurred_at	merchant_country_code	merchant_city	latitude	longitude	description	search_vector

risk_rule

rule_id	rule_code	rule_name	dimension	weight	threshold_value	is_active	version

risk_signal

signal_id	transaction_id	rule_id	observed_value	score_contribution	evidence	triggered_at

fraud_scenario

scenario_id	scenario_code	scenario_name	description

transaction_label

transaction_id	scenario_id	is_suspicious

investigation_case

case_id	case_ref	transaction_id	assigned_to_user_id	status	priority	resolution_code	resolution_notes	opened_at	due_at	closed_at	updated_at	version

case_event

event_id	case_id	actor_user_id	event_type	from_state	to_state	reason	metadata	operation_id	event_ordinal	occurred_at

case_command_receipt

operation_id	actor_user_id	idempotency_key	command_type	request_hash	case_id	response_status	completed_at

app_session

session_id	user_id	created_at	last_seen_at	idle_expires_at	absolute_expires_at	revoked_at	revocation_reason

app_session_cleanup

operation_id	admin_id	preview_at	cutoff_at	expires_at	candidate_session_ids	candidate_count	status	idempotency_key	request_hash	completed_at	deleted_count

schema_migrations

version	filename	checksum	applied_at

migration bookkeeping

v_transaction_risk

transaction_id	risk_score	signal_count	risk_band

view columns; no declared primary key

device_telemetry [Mongo collection]

_id	device_id	transaction_id	captured_at	schema_version	shape_type	retention_eligible	payload

payload when shape_type = connection_telemetry

ip_routing_hash	connection_type	latency_ms	tls_cipher	vpn_detected

payload when shape_type = client_environment

screen_resolution	battery_level	battery_charging	os_patch_level	hardware_concurrency	user_agent_category

Solid underline = PK. Dotted underline = FK (a field can be both). Red arrows = PK determines other columns. Black arrows = declared SQL FK references.

Cream cells = stored generated columns. Composite FKs: transaction(account_id,currency_code) -> account(account_id,currency_code); transaction(method_id,account_id) -> payment_method(method_id,account_id).

Mongo device_id/transaction_id are application-validated references, not SQL FKs; payload is one of the two shapes. No 3NF claim is made for derived fields, the view or document store.

Source: main 843f0af95974 | migrations 010-060 | 30 October 2026 | A1 landscape, vector